Counting configurations above a cyclic subgroup of order three

Human Compression Project

Abstract

One regular action of a dihedral quotient counts all configurations whose maximal parabolic is A_4 and whose intersection has order three. Its reflections describe the second parabolics S_3, while its nonidentity elements describe ordered pairs of A_4 parabolics. Passing to unordered pairs identifies an element with its inverse. This recovers five rows of the source table and corrects the conjugacy counts in its last two rows. The full automorphism counts in those rows remain valid.

1 The hypotheses and the common normalizer

Let $p > 3$ be prime such that A_4 is maximal in $G = \mathrm{PSL}(2, p)$. The classical maximal-subgroup list gives exactly $p = 5$ or $p \equiv 3, 13, 27, 37 \pmod{40}$. In particular $p \equiv \pm 3 \pmod 8$. Put $E = \mathrm{PGL}(2, p) = \mathrm{Aut}(G)$, and choose $\epsilon \in \{1, -1\}$ with $3 \mid p - \epsilon$. Write

$$n = \frac{p - \epsilon}{2} = 3m, \qquad m = \frac{p - \epsilon}{6}.$$

Fix $B \cong C_3$ in G. We retain the classical facts that all such B are conjugate, that the subgroups A_4 form one G-class, and that $\mathrm{N}_E(A_4) \cong S_4$. The latter group acts on its normal A_4 by the usual automorphisms. Also, the centralizer torus of B has order n in G and $2n$ in E, and its normalizers are dihedral. These are subgroup facts, not assumptions about the number of geometries.

Choose torus and reflection generators so that

$$\mathrm{N}_E(B) = \langle R, S \mid R^{2n} = S^2 = 1,\ SRS = R^{-1} \rangle, \quad \mathrm{N}_G(B) = \langle R^2, S \rangle, \quad B = \langle R^{2m} \rangle.$$

After quotienting by B, write

$$\overline{N} = \mathrm{N}_E(B)/B = \langle x, s \mid x^{2m} = s^2 = 1,\ sxs = x^{-1} \rangle, \qquad K = \mathrm{N}_G(B)/B = \langle r = x^2, s \rangle.$$

Thus $|K| = 2m$ and r has order m. This presentation also covers $m = 1, 2$ without an exceptional convention for a small dihedral group.

Lemma 1. *The set $\mathcal{A}$ of subgroups A_4 containing B is a regular K-set. Choose one of them, denoted M, as the identity point. The stabilizer of M in $\overline{N}$ is generated by an external reflection j. Coordinates may be chosen with $j = xs$; its action on the K-coordinates is the automorphism*

$$\theta(r^k) = r^{-k}, \qquad \theta(r^k s) = r^{1-k} s.$$

Proof. All subgroups C_3 in A_4 are conjugate internally, and their normalizer in A_4 is C_3. Any transporter between two members of $\mathcal{A}$ can therefore be adjusted inside the target A_4 to normalize B. Hence $\mathrm{N}_G(B)$ is transitive on $\mathcal{A}$. Since M is maximal in the simple group G, it is self-normalizing. The stabilizer of M is consequently $\mathrm{N}_G(B) \cap M = \mathrm{N}_M(B) = B$. Moreover B

normalizes every member of $\mathcal{A}$, since it is contained in each. The action therefore factors through K and is regular.

The stabilizer in $\mathrm{N}_E(B)$ is $\mathrm{N}_{\mathrm{N}_E(M)}(B)$, the normalizer of a three-cycle subgroup in S_4. It is S_3, with quotient C_2 by B. Its involution inverts B, so its image in $\overline{N}$ is a reflection. It is outside K, since the stabilizer in K is trivial. Every external reflection has the form $x^{2a+1}s$; changing the initial reflection generator permits $j = xs$. Conjugation by j sends r to r^{-1} and s to rs. Since j fixes the identity point M in the regular set, these conjugation formulas are exactly its action on the coordinates. □

2 Different parabolic types

The common overgroup calculation used in the source audit has the following interface: if $H \leq G$ properly contains B and its action on H/B is two-transitive, then H is C_6, S_3 or A_4. For this particular B the list can also be checked directly. Writing $d = [H : B]$, transitivity of B on the other $d - 1$ cosets gives $d - 1 \mid 3$, so $d = 2$ or 4. At degree two H has order six and is cyclic or symmetric. At degree four the kernel, being contained in B, has order one or three. A kernel of order three would leave a group of order four acting two-transitively on four points, impossible. Thus the action is faithful and its image is the unique subgroup A_4 of order twelve in S_4. This proves the interface without an additional finite search.

Theorem 2. *The configurations with parabolics $M \cong A_4$ and $H \cong C_6$ exist exactly when m is even; then there is one conjugacy class and one E-class. With $H \cong S_3$, there are m conjugacy classes and $\lceil m/2 \rceil$ classes under E.*

Proof. A cyclic overgroup C_6 must lie in the unique torus centralizing B. It exists exactly when $6 \mid n$, equivalently m even, and is then unique. Fixing M and its subgroup B loses no configuration because both choices form a single chain class. This proves the first count.

A subgroup S_3 containing B normalizes B and is the inverse image of a reflection subgroup of order two in K. Conversely each reflection $r^k s$, with $k \in \mathbb{Z}/m\mathbb{Z}$, gives one such S_3. Indeed its inverse image has order six and inverts B. There are exactly m possibilities. The stabilizer of the chain (M, B) inside G is B, which fixes every one of these overgroups. Thus all m give distinct conjugacy classes of configurations. For the E-classes we further quotient by j, whose action is $k \mapsto 1 - k$. This involution has one fixed point for odd m and none for even m, proving the count $\lceil m/2 \rceil$.

In both types, the intersection with M is exactly B: B is maximal in M and M cannot lie in a group of order six. Neither group of order six is a subgroup of A_4, so M and H generate G by maximality of M. Their local actions are respectively the natural action of A_4 of degree four and an action of degree two. This verifies the rank two geometry and local two-transitivity conditions, rather than just counting abstract pairs. □

3 Equal parabolic types and inversion

Lemma 3. *If a group K acts regularly on a set, its orbits on ordered pairs of distinct points are parametrized by $K \setminus \{1\}$. Its orbits on unordered pairs are parametrized by the relation $k \sim k^{-1}$ on that same set.*

Proof. Using the regular coordinates, a pair (a, b) has invariant $a^{-1}b$. Translate its first entry to the identity to see that this invariant is complete. Reversing the pair replaces the invariant by its inverse. These are exactly the further identifications when order is forgotten. □

Theorem 4. *Under the source's unordered convention, the configurations with two parabolics A_4 meeting in B have*

$$m + \left\lfloor \frac{m}{2} \right\rfloor \quad \textit{conjugacy classes, and} \quad m \quad \textit{classes under } E.$$

Proof. Two distinct members of $\mathcal{A}$ meet in exactly B, since B is maximal in each A_4. They generate G, and both local actions have degree four and are two-transitive. Conversely every configuration of the specified type occurs this way. Any conjugacy between two such pairs normalizes their intersection B. The conjugacy problem is therefore precisely the regular action of K on unordered pairs in $\mathcal{A}$.

All m reflections in K are self-inverse. Among the nonidentity rotations, inversion has $\lfloor m/2 \rfloor$ orbits. Lemmas 1 and 3 give the asserted inner count. For E, the additional relation is $k \sim \theta(k)$. On rotations this already is inversion. On reflections inversion is trivial, whereas θ is $k \mapsto 1 - k$ on their exponents. Thus the number of classes under E is

$$\left\lfloor \frac{m}{2} \right\rfloor + \left\lceil \frac{m}{2} \right\rceil = m.$$

This also shows exactly why the source's full automorphism count survives the correction to its conjugacy count. □

4 The two source passages

Table 2 of the published Part II article has seven rows. Its first row is Theorem 2 for C_6; combining m even with the maximal-A_4 congruences gives exactly $p \equiv 13, 37, 83, 107 \pmod{120}$. Rows 2–5 are the S_3 case, separated by the sign of $p - \epsilon$ and the parity of m. All five rows retain their two class counts.

Rows 6–7 concern two A_4 parabolics. They print the inner count $(p-\epsilon)/3-1 = 2m-1$, which counts ordered pairs in the regular set. Section 2 of the published source defines configurations using unordered sets of parabolics. Their correct inner count under that definition is $m + \lfloor m/2 \rfloor$. The printed full automorphism count $(p-\epsilon)/6 = m$ is recovered. For example, the inner count at $p = 37$ is nine instead of eleven, and at $p = 43$ it is ten instead of thirteen. These are consequences of the uniform proof, not numerical searches.

The fixed appendix repeats the same calculation in Proposition 16, Subcase 3, on its physical page 4. It subtracts one from the regular set size but does not then divide by inversion. The same correction therefore disposes of that count in the second original. Its Subcases 1–2, physical pages 2–4, have the recovered different-type counts proved above. The exceptional case $p = 5$ has $m = 1$ and both counts equal one, so the correction does not alter that row's value.

The original configurations and the classical subgroup premises are attributed to the source and its cited subgroup classifications. The common normalizer description, explicit reflection action and inversion quotient are the derivation used here. This component does not complete either original, add a principal refutation, claim a new open-conjecture answer, claim Lean verification or measure compression.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Section 2, Table 2 and Proposition 5.6.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition, Proposition 16, physical pp. 2–4.